

A versatile computer-controlled pulsed nuclear quadrupole resonance spectrometer

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A versatile computer-controlled pulsed nuclear quadrupole resonance spectrometer

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A new, pulsed nuclear quadrupole resonance (NQR) spectrometer capable of performing a variety of pulsed and swept experiments is described. The spectrometer features phase locked, superheterodyne detection using a commercial spectrum analyzer and a fully automatic, computer-controlled tuning and matching network. The tuning and matching network employs stepper motors which turn high power air gap capacitors in a “moving grid” optimization strategy to minimize the reflected power from a directional coupler. In the duplexer circuit, digitally controlled relays are used to switch different lengths of coax cable appropriate for the different radio frequencies. A home-built pulse programmer card controls the timing of radio frequency pulses sent to the probe, while data acquisition and control software is written in Microsoft Quick Basic. Spin-echo acquisition experiments are typically used to acquire the data, although a variety of pulse sequences can be employed. Scan times range from one to several hours depending upon the step resolution and the spectral range required for each experiment. Pure NQR spectra of NaNO_2 and 3-aminopyridine are discussed. © 1999 American Institute of Physics. [S0034-6748(99)00912-0]

I. INTRODUCTION

Nuclear quadrupole resonance (NQR) spectroscopy has undergone significant development in recent years. This is primarily due to the interest in investigating quadrupolar nuclei for which nuclear magnetic resonance (NMR) is currently incapable of observing, as well as advances in NQR instrumentation. The development of instrumentation for detection of pure quadrupolar resonance in solids has paralleled developments made in NMR.^{1,2} Traditionally, experimental techniques for determining NQR spectra have been divided into three main categories: (1) continuous-wave (cw) methods that employ regenerative^{3,4} or super-regenerative oscillators,⁵ (2) pulsed techniques,^{6,7} and (3) field cycling techniques.^{8,9} Although a number of problems originating primarily from stability concerns were often associated with many of the early cw techniques, they did offer the ability to scan across wide spectral ranges and thus provided much of the early information about quadrupolar nuclei and quadrupole coupling constants.¹⁰

With the advent of pulsed methodologies, many of the problems associated with cw experimental techniques have been overcome. Numerous pulsed NQR spectrometers have been constructed and reported over the last twenty years.^{11–15} The advantages of pulsed excitation include: (1) stable operation using linear circuitry, (2) avoidance of saturation effects inherent in cw methods, (3) the ability to conveniently measure relaxation times, and (4) strategies for signal averaging and performing other data analyses. For quantitative analysis, in particular, when the resonance frequency of interest is known in advance, NQR can provide a unique strategy as has been reported for drug and explosives

detection.^{16–18} Although significant progress has been made with NQR instrumentation, the technique still suffers from other limitations. In the more general case, the broad range of quadrupole coupling constants observed for a given nucleus make locating an unknown resonance challenging. For example, the frequency range for ^{14}N NQR in nitrogen containing compounds is several MHz, whereas the linewidths are typically a few kHz. Pulsed amplifiers are not available which can cause effective mixing of states across such a broad spectral range. Recently, so-called “noise excitation” schemes employing short pulses and correlation techniques for signal detection have shown significantly better excitation profiles.¹⁹ Nevertheless, the spectral width of the measurement is most often determined by the resonant frequency range or “Q” of the probe and the limitations imposed by radio-frequency (rf) filter and mixer stages, which are still smaller than the typical range of most NQR observable nuclei. As a result, the resonant probe circuits used originally for pulsed NQR experiments had to be manually tuned to observe a specific transition at a given frequency during the experiment. Though useful for relaxation measurements and for studying compounds whose coupling constants had been previously determined, pulsed NQR spectrometers were not originally used to search for quadrupolar coupling constants in new compounds or materials.

The incorporation of super-heterodyne detection schemes and quadrature phase sensitive detection with these “swept” pulsed experiments by Marino and co-workers were the first attempts to combine modern circuitry for NQR detection.¹⁴ Efforts to combine the advantages from both cw and pulsed methods have now been undertaken by various research groups. More recently, some of these instruments are now computer-controlled/fully automated, auto-tuned NQR spectrometers.^{14,20} These spectrometers employ banks

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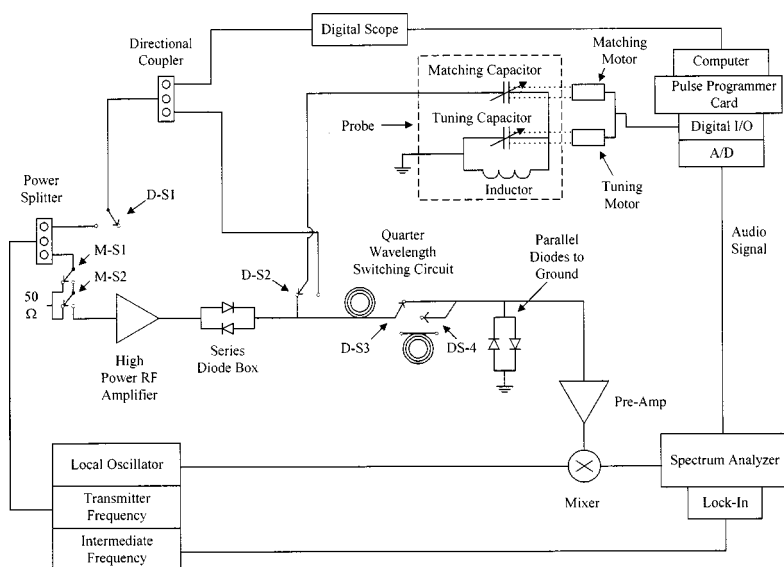


FIG. 1. Schematic diagram of the NQR spectrometer. The spectrometer features phase locked, superheterodyne detection using a commercial spectrum analyzer and a fully automatic, computer-controlled tuning and matching network. Details are given in the text.

of capacitors and/or inductors which are switched in at predetermined intervals during the NQR experiment to tune the LC tank circuit. Another significant development in NQR spectrometer construction is the incorporation of digitally controlled frequency synthesizers that allow the generation of pulses of high spectral purity and arbitrary phase.²¹ Such experimental control is often essential for developing multiple pulse sequences that require high time resolution and fast update times for phase cycling. Additionally, commercially available software packages such as LABVIEW have been implemented to control the NQR experiment and can help reduce the time consuming task of software development.

Many of these spectrometers have been used to look at moderate to lower frequency quadrupolar nuclei such as ^{35}Cl and ^{14}N . For quadrupolar nuclei such as ^{23}Na , ^2D , and ^{27}Al , whose coupling constants are often below 600 kHz, NQR detection is often difficult. Experiments involving these nuclei are therefore typically performed in high magnetic field. Researchers have developed field cycling techniques,^{22,23} and have even employed superconducting quantum interference device dc (SQUID) detectors²⁴ to provide the sensitivity necessary to observe these weak NQR transitions. Other researchers have modified pre-existing NMR spectrometers to observe quadrupole transitions of higher frequency nuclei such as ^{79}Br and ^{81}Br .²⁵ NQR detection of these nuclei often employ loop-gap resonator circuits as opposed to conventional solenoid circuitry.²⁶ A recent article combines an auto-tuning procedure with loop-gap resonator circuitry to obtain ^{79}Br and ^{81}Br spectra of flame retardants.²⁵

The spectrometer we describe combines many of the advantages detailed above with additional improvements made in the tuning and matching routine for doing swept NQR experiments. The spectrometer employs a spectrum analyzer as the receiver, which, in addition to considerably simplifying rf design and construction, provides a useful tool for performing diagnostic tests on the instrument as necessary. To date, experiments to detect nuclei at resonant frequencies from 300 kHz to 40 MHz have been carried out with this instrument. A description of the spectrometer is provided

along with some of the experiments that have been performed to demonstrate its versatility.

II. INSTRUMENT DESCRIPTION

The spectrometer features phase locked, superheterodyne detection using a commercially available spectrum analyzer (Hewlett Packard, containing HP8552B-IF, HP8553B-RF, and HP8443A Tracking Generator module) and a fully automatic, computer-controlled tuning and matching network. All frequencies, the local oscillator (LO) and transmitter, as well as reflected power measurements, are controlled through a general purpose interface bus (GPIB). A schematic diagram of the spectrometer is shown in Fig. 1.

A. Pulse programmer

The pulse programmer was built on an IBM PC-compatible printed circuit card and consists of four Signetics Corp. 7202 first in, first out (FIFO) chips, each of which contains 1 k 8 bit words, along with a 24 bit down counter, control logic and an 8 bit data output register. A pulse program consists of a series of 32 bit words, each of which consists of an 8 bit data word used to control the rf gates and data acquisition trigger, and an associated 24 bit timing word. The pulse program, consisting of user-defined commands such as "pulse" and "delay," that are defined in the spectrometer software, is loaded sequentially onto the FIFO. The data and timing words are loaded through the IBM PC Bus onto the FIFO using separate, but sequential IO addresses. Once the pulse program is loaded, a "go" command starts the pulse programmer, which immediately loads the first data word onto the output register. This data word stays on the output register until the associated timing word counts from its initial value down to zero in 1 μs steps. Additional data words are then loaded onto the output register until the pulse program is complete. The 24 bit timing word allows pulses or delays from 1 to 2^{24} μs (16.8 s). Although simple by present commercial standards, this home built pulse programmer is quite versatile and very cost effective. Pulse programs are relatively easy to write using the software. As

currently configured, a pulse program is written as a subroutine in Quick Basic that must be separately compiled. It would not be difficult, however, to implement a pulse program file that was read by the once-compiled main program. Additional information on the pulse programmer card and/or spectrometer software is available from the authors.

B. Transmitter

Pulses are generated in the following way: The transmitter frequency is set on a SRS synthesizer (Stanford Research Systems, DS345) via a GPIB card, the output is routed through a power splitter (Mini-Circuits, ZMSC-2-1) into M-S1 and M-S2 (Mini-Circuits, ZYSWA-2-50DR), and then on to a high power rf amplifier (ENI Model 350L). M-S1 and M-S2 are fast rf switches controlled by the pulse programmer card that supply the transmitter pulses to the rf amplifier. The high power rf is now routed through D-S2 (Dow-Key-401 Series, SPDT) and then to the probe. Switch D-S2, shown in the normally closed position, determines whether the pulsed NQR experiment or the tuning and matching routine is to be performed.

C. Probe

The probes used in these experiments consist of an “N” turn, solenoid-type inductor, and two tunable (13.5–300 pF), high power, air-gap capacitors (Caldwell Condensor Corp). The inductor is made from 22 gauge insulated copper wire and is wrapped and epoxied to a Delrin coil form. A 10 cc Pyrex vial is placed in the coil form to hold the samples. To observe other nuclei, the size and number of turns of the inductor were modified. Shielding of the signal path is accomplished by placing the capacitors in an aluminum chassis (10 in. $\times$ 20 in.) and the inductor in a brass tube (4 in. diam., 8 in. tall, 1/4 in. wall thickness). The mechanical coupling between the capacitors and the stepper motors is made with a nonconducting piece of G-10 fiberglass rod to ensure that there is no electrical contact between the stepper motors and the LC tank circuit. Doubly insulated 50 Ω coaxial cable (Belden 9907) was used between the probe and the input to the analog-to-digital (A/D) card to provide additional rf isolation.

D. Duplexer

After the probe, the signal path is routed through a $\lambda/4$ duplexer circuit. Due to the large frequency range typically employed in most swept NQR experiments, a single $\lambda/4$ cable does not adequately protect the preamp over the whole tuning range. ^{14}N NQR is particularly difficult in this regard, since typical ^{14}N NQR frequencies range from 2 to 5 MHz. Switches D-S3 and D-S4 (Dow-Key 401 Series, SPDT) are used to add additional lengths of cable at predetermined times during the experiment. These switches, as well as switch D-S2, are positive-true logic, TTL compatible, and are accessed through digital I/O lines by the computer program. For frequencies between 3 and 5 MHz, switches D-S3 and D-S4 are placed in their open positions (as shown in Fig. 1). In this state, the signal path is through the first $\lambda/4$ section, a 38.8 ft length of cable which is a $\lambda/4$ length at 3.8

MHz. At frequencies between 2 and 3 MHz, switches D-S3 and D-S4 are placed in their normally closed position. In this state, the signal path is through both the first and second $\lambda/4$ sections. The second $\lambda/4$ is an additional 22 ft cable and serves to place the $\lambda/4$ of the circuit at 2.4 MHz. The $\lambda/4$ cable lengths are chosen to provide adequate protection to the preamp from overload. At the same time, they do not reduce the power delivered to the probe by an unacceptable amount for transmitter frequencies within 25% of the actual quarter wavelength in the circuit. Additional $\lambda/4$ lengths could be added for larger frequency sweeps.

E. Preamp and receiver

A home-built, 30 dB gain preamp is employed to increase the signal amplitude sufficiently before it is routed to the spectrum analyzer. The preamp design incorporates three MAR4 (Mini-Circuits) amplifiers. These amplifiers offer flat broadband gain from dc to 2 GHz with a typical noise figure (NF) of 3.5 dB. A single transistor stage preamplifier precedes the MAR-4 amplifiers to reduce the NF further. The NF of the receiver system was measured to be roughly 2 dB using a 50 Ω load and measuring the noise voltage at two temperatures.²⁷ After the preamp, the signal is mixed with the local oscillator (LO) through a mixer (Mini-Circuits, ZAD-1) up to an intermediate frequency (IF), which is typically 10.5 MHz, and is then routed to the spectrum analyzer.

The HP spectrum analyzer was used as the heart of the receiver system as it provides a versatile platform with which to construct a convenient, yet broadband and multipurpose receiver. One advantage of using the spectrum analyzer as the receiver is that the IF frequency can be varied from 100 kHz to 110 MHz. The receiver system (including both the preamp and spectrum analyzer) has a gain of 100 dB from 1 to 100 MHz. The use of an HP spectrum analyzer as a triple conversion receiver for broadband NMR detection has been described previously.²⁸ Briefly, the signal derived from mixing the rf signal from the preamp output with the LO frequency is applied to the spectrum analyzer's rf section (HP module 8553B-RF), where it is first amplified by 30 dB and then converted to a 200 MHz signal. Up conversion of the signal reduces spurious responses from harmonic mixing signals. The 200 MHz carrier is then converted to a 50 MHz IF signal and then to 3 MHz using LO frequency sources (150 and 47 MHz) that are generated by voltage controlled oscillators located within the spectrum analyzer. The 3 MHz section contains a series of phase-corrected crystal and LC filters that can be used to reduce the bandwidth from 300 kHz to 10 Hz (HP module 8552B-IF). Normally the filter bandwidth was set to 10 kHz. The 3 MHz signal is finally down converted to audio frequencies externally, and this output is fed to the A/C converter for detection.

To maintain phase coherence in the receiver system, a home-built lock-in amplifier is employed, which consists of a wideband mixer that compares the IF frequency with that of the output from the 8443A Tracking Generator. The dc output at zero beat is then fed back into the scan control input of the 8552B-IF section to provide external control of the local oscillators in the spectrum analyzer. The feedback loop in-

incorporates a buffer amplifier which limits the frequency lock response from dc to 50 kHz and is designed for high stability. Because the feedback loop operates at the IF frequency (10.5 MHz typically), phase coherence between the transmitted carrier frequency and the receiver system is maintained.

F. Tuning and matching

In order to tune and match the probe, low level rf (5 dBm typical) is routed through the power splitter, through switches D-S1 and D-S2 in their normally open positions, and into the directional coupler (Mini-Circuits, ZMDC-10-1). The magnitude of the reflected power from the probe is measured with a digital oscilloscope and monitored through its GPIB interface by the computer. Alternatively, the signal can be amplified and diode detected to reduce cost. For each measurement, the program compares the reflected power of the probe circuit with a predetermined maximum acceptable value. The magnitude of the acceptable reflected power varies depending upon the input tuning level. For these experiments, a reflected power level corresponding to a V_{pp} of 70 mV was predetermined to not affect the signal-to-noise (S/N) of the NQR measurement adversely. If V_{pp} is less than 70 mV, the circuit is considered properly tuned and matched to 50 Ω . If V_{pp} is greater than 70 mV, the computer sends a TTL signal through a digital I/O port to a microstep translator/drive (SLO-SYN, SS2000MD4-M). The translator/drive controls the movement of two tuning and matching stepper motors (SLO-SYN, M062-FF-206) in a “moving grid” minimization routine. The total grid size is determined by the step resolution that has been set on the translator drives with each position in the grid corresponding to a unique combination of different tuning and matching motor positions. The step resolution used in these experiments provided a total grid of 200 $\times$ 200. The minimization routine works as follows: (1) The positions of the tuning and matching motors are incremented by one step in an iterative manner around a predetermined “subgrid” (typically 10 $\times$ 10) of the total grid. The reflected power at each increment is measured and compared to the threshold value of 70 mV_{pp}. If the reflected power at any position in the subgrid is less than the threshold value, the motors moved to that position and the pulsed experiment is resumed. (2) If the condition for the threshold minimum is not found within the 10 $\times$ 10 subgrid, the tuning and matching motors are moved to the position in the subgrid where the lowest amount of reflected power was measured. This position is now considered the starting point. (3) The process of moving the stepper motors around the new subgrid is repeated and is continued until a reflected power less than the threshold is found. Once the circuit is properly tuned, switches D-S1 and D-S2 are moved back to the normally closed position and the pulsed experiment is resumed. To reduce the chance of electrical noise problems, the microstep translator/drives use optical isolation between the input control signals and the translator drive circuitry. A driver circuit consisting of a combination 4049, hex inverting buffer/driver and a 7406, hex inverter buffer/driver with open collector high-voltage output was used to interface the computer digital I/O to the TTL inputs of the microstep

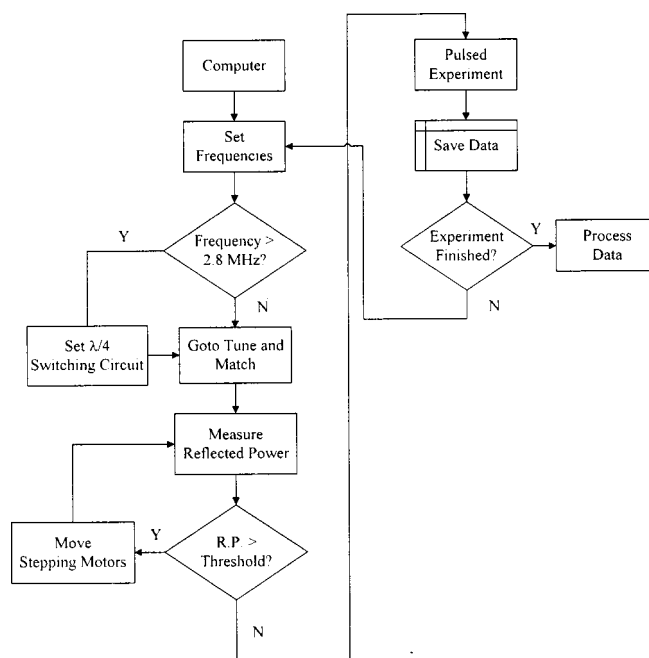


FIG. 2. Flow diagram showing the control scheme for the swept NQR experiment.

translator/drives to provide sufficient input drive current (16 mA).

The experimental parameters for the sample, the beginning and ending frequencies, the step size of the search, and other pertinent NQR parameters such as delays, pulse widths, etc. are chosen at the beginning of the experiment. The software as written allows several routine NQR experiments to be performed, including single pulse/acquire, Hahn echo ($\pi/2$ - τ - π - τ -acquire), and presaturation/Hahn echo [$(\pi/2$ -delay)_n- τ_1 - $\pi/2$ - τ_2 - π - τ_3 -acquire] for T_1 determinations. A Hahn echo pulse sequence was employed in the experiments described below. Figure 2 shows a flow diagram of the swept experiment.

G. Data processing

For the swept experiment we evaluated a number of data processing methods. The following method gave the best results: At the completion of each experiment the free induction decay (FID) at each frequency step was Fourier transformed and the amplitude of the largest peak in the frequency spectrum was determined. The amplitude of this peak was divided by the average of the last 50 data points in the time domain data. These 50 data points constitute an estimate of the noise in the spectrum. A ratio of the largest signal amplitude to the noise is calculated and is saved at each frequency increment. The S/N ratio at each frequency increment was combined to give the resultant spectrum. This spectrum can then be displayed and or saved as a text file. These data can then be further examined or further processed using a program written in QuickBasic. Alternatively, they can be imported into a commercial spreadsheet. Typically the individual data files containing the FID information were

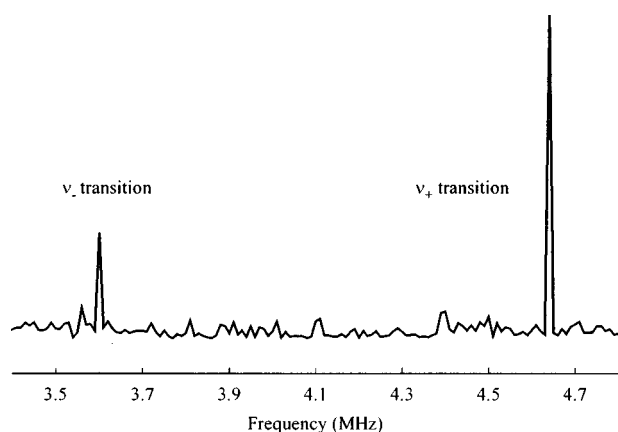


FIG. 3. Room temperature spectrum showing the ν_+ and ν_- transitions in NaNO_2 . The spectrum was recorded with a 10 kHz step size from 3.40 to 4.80 MHz. Acquisition time for the spectrum was approximately 5 h.

saved to allow further examination. Often, additional NQR experiments were performed at frequencies of interest as identified using the swept experimental spectrum.

III. RESULTS AND DISCUSSION

A. Data from swept experiment

The capabilities of the instrument were examined using samples of NaNO_2 and 3-aminopyridine. NaNO_2 is well characterized in terms of its NQR parameters²⁹ and has short relaxation times at room temperature making it well suited for preliminary studies. Transition frequencies for 3-aminopyridine have also been previously reported in the literature³⁰ but this compound provided an additional test sample for which four NQR transitions should be observed in a relatively small tuning range. The tuning range of our ^{14}N probe with a twenty-turn inductor is 2.3–4.8 MHz. This entire range can be investigated without having to make any modifications to the hardware and includes most nitrogen coupling constants of interest.

Figure 3 shows the room temperature spectrum of the NaNO_2 . The spectrum was recorded with a 10 kHz step size from 3.40 to 4.80 MHz. Fifty transients were taken at each frequency step with a recycle delay of 1 s. The total acquisition time of the spectrum was approximately 5 h. The spectrum shows two well-resolved lines at approximately 3.60 and 4.65 MHz. Fourier transformation of the individual FIDs acquired at these frequencies indicated that the ν_- resonance occurs at 3.603 MHz, while the ν_+ resonance appears at 4.635 MHz.

Figure 4 shows the room temperature spectrum of 3-aminopyridine. The two transitions for the amino nitrogen were observed at 2.451 and 3.131 MHz and the two transitions for the ring nitrogen were located at 2.89 and 3.72 MHz. Data acquisition was from 2.35 to 3.82 MHz in 10 kHz increments. For this experiment, 50 transients were taken at each frequency step with a recycle delay of 5 s. Approximately 10 g of sample was used in both experiments.

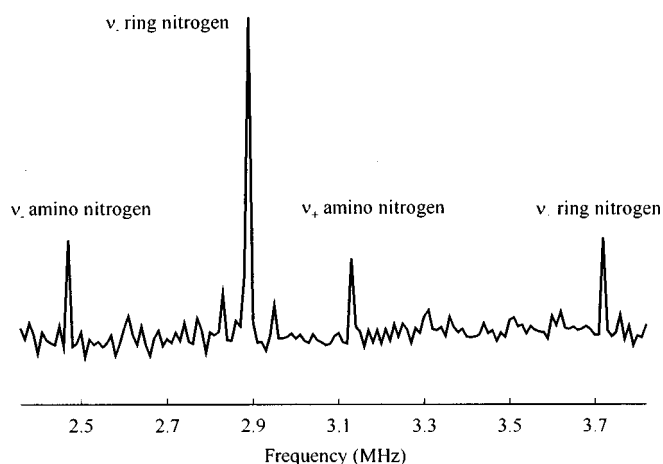


FIG. 4. Room-temperature spectrum showing the ν_+ and ν_- transitions for the amino and ring nitrogens in 3-aminopyridine. Data acquisition was from 2.35 to 3.82 MHz in 10 kHz increments. Acquisition time for the spectrum was approximately 6 h.

B. Data from pulsed experiments

Once a swept experiment has been completed, the spectrometer can be easily converted by software to function as a single frequency pulsed NQR spectrometer. Figure 5 shows representative data from the time and frequency domain of the ν_+ transition in NaNO_2 from a pulsed NQR experiment. The time domain spectrum is the result of 50 transients using a Hahn echo pulse sequence. A 24 μs , 70 W pulse was used to excite the sample, along with a recycle delay (D1) of 1 s and τ values in the 2 ms range. The frequency spectrum shows a single line that appears at 4.6346 MHz (4.6 kHz above the 4.630 MHz carrier frequency). The NaNO_2 resonances have significant temperature dependences which account for small changes in the observed frequencies.²⁹ The observed linewidth of approximately 1200 Hz and S/N ratio of 100:1 is comparable to results reported by Bray and co-workers.²⁹ Pure NQR Cl transitions in NaClO_3 at 29.935 MHz and the ^1H resonance in a water sample at 300 kHz in a home-built solenoid magnet (not shown here) have also been observed with this spectrometer.

We also performed a series of T_1 experiments on a va-

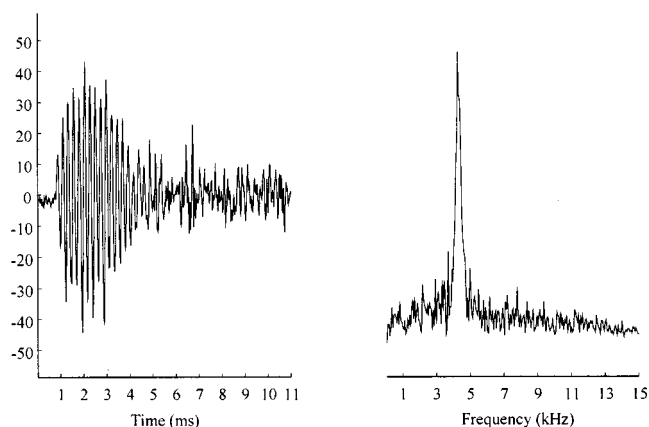


FIG. 5. Time and frequency domain data for the ν_- transition in NaNO_2 obtained with the NQR spectrometer configured for pulsed data acquisition. The time domain data is the result of 50 transients using a Hahn echo pulse sequence.

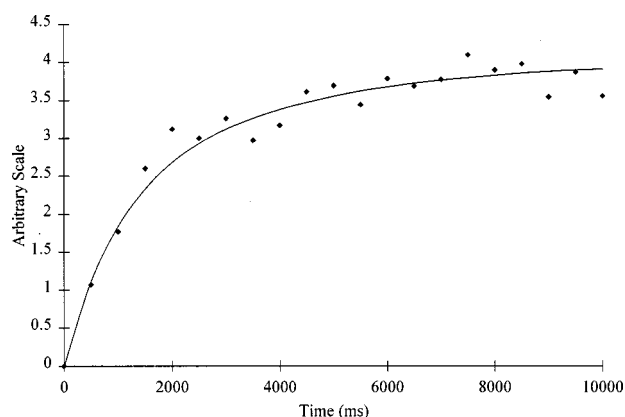


FIG. 6. Plot of experimental T_1 data for the ν_+ transition of the amino nitrogen in 3-aminopyridine along with a best fit line that was determined using a χ -square grid search. The solid line represents the best fit using T_1 values of 930 and 3300 ms. Experimental data are represented by individual points. Experimental details are given in the text.

riety of nitrogen containing compounds using a presaturation pulse sequence. Figure 6 shows experimental T_1 data for the ν_+ transition of the amino nitrogen in 3-aminopyridine along with a best fit line that was determined with a χ -square grid search routine. The pulse sequence consisted of a train of thirty $15\ \mu\text{s}$ presaturation pulses separated by 25 ms, followed by a τ_1 increment and a Hahn echo pulse sequence. τ_1 values were incremented from 0 to 10 s in 500 ms steps. A $15\ \mu\text{s}$, 70 W $\pi/2$ pulse was used along with signal averaging of 100 transients at each increment. Relaxation in spin 1 nuclei is known to follow a rate equation that is the result of the sum of two independent relaxation pathways.²⁹ The two best fit T_1 values determined in this experiment were 930 and 3300 ms.

IV. CONCLUSIONS

A new, fully automated, computer-controlled NQR spectrometer is demonstrated which is capable of performing a variety of different NQR experiments. The instrument can be easily modified to study other nuclei by changing the size of the inductor in the probe circuit and the lengths of the $\lambda/4$ cables. The use of a spectrum analyzer, which can be obtained at reasonable cost, simplifies rf design and construction. The multistage receiver system utilized by the spectrum analyzer offers both signal gain and noise filtering at three different IF stages which give the system the sensitivity necessary for detecting low level NQR signals.

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